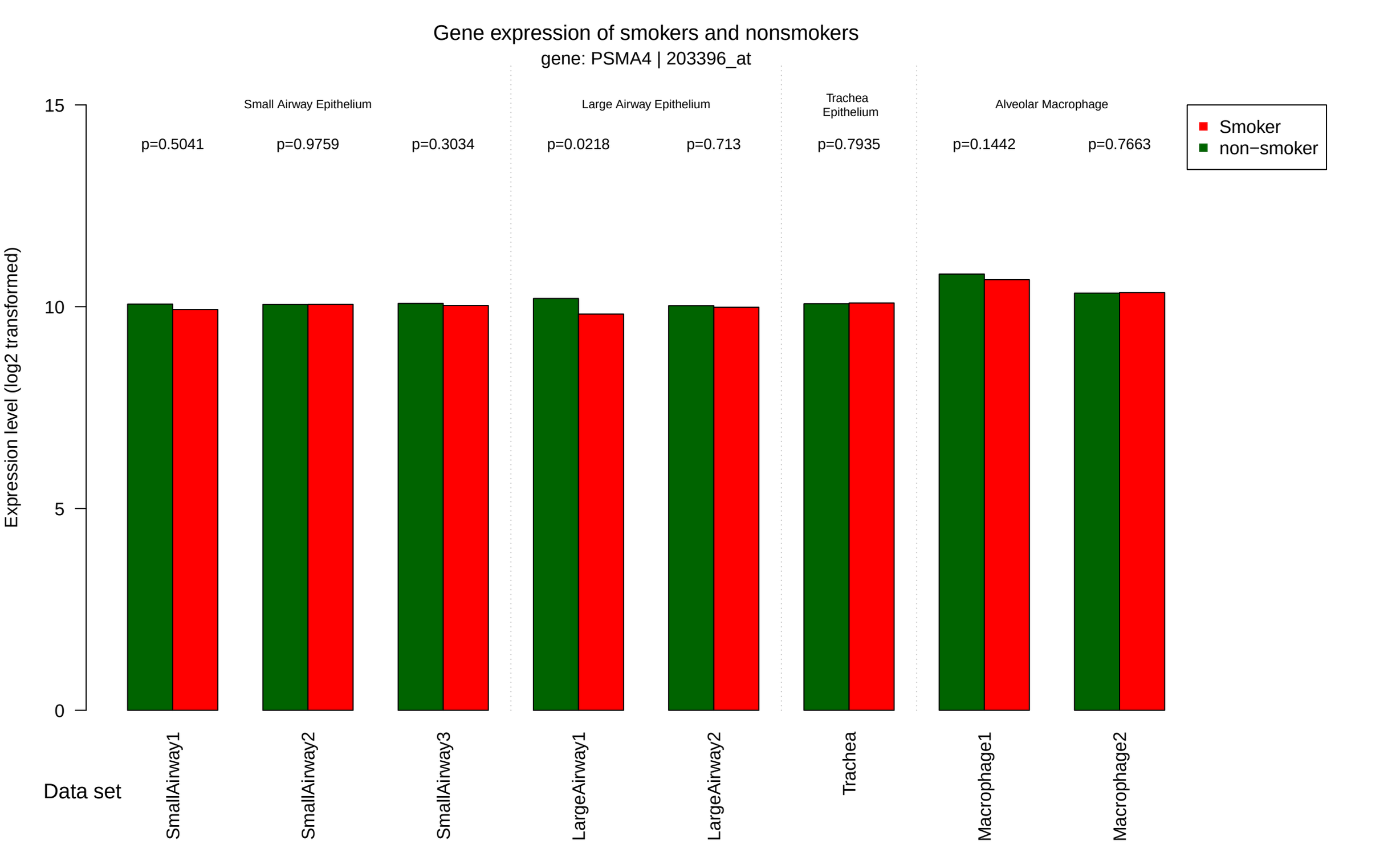
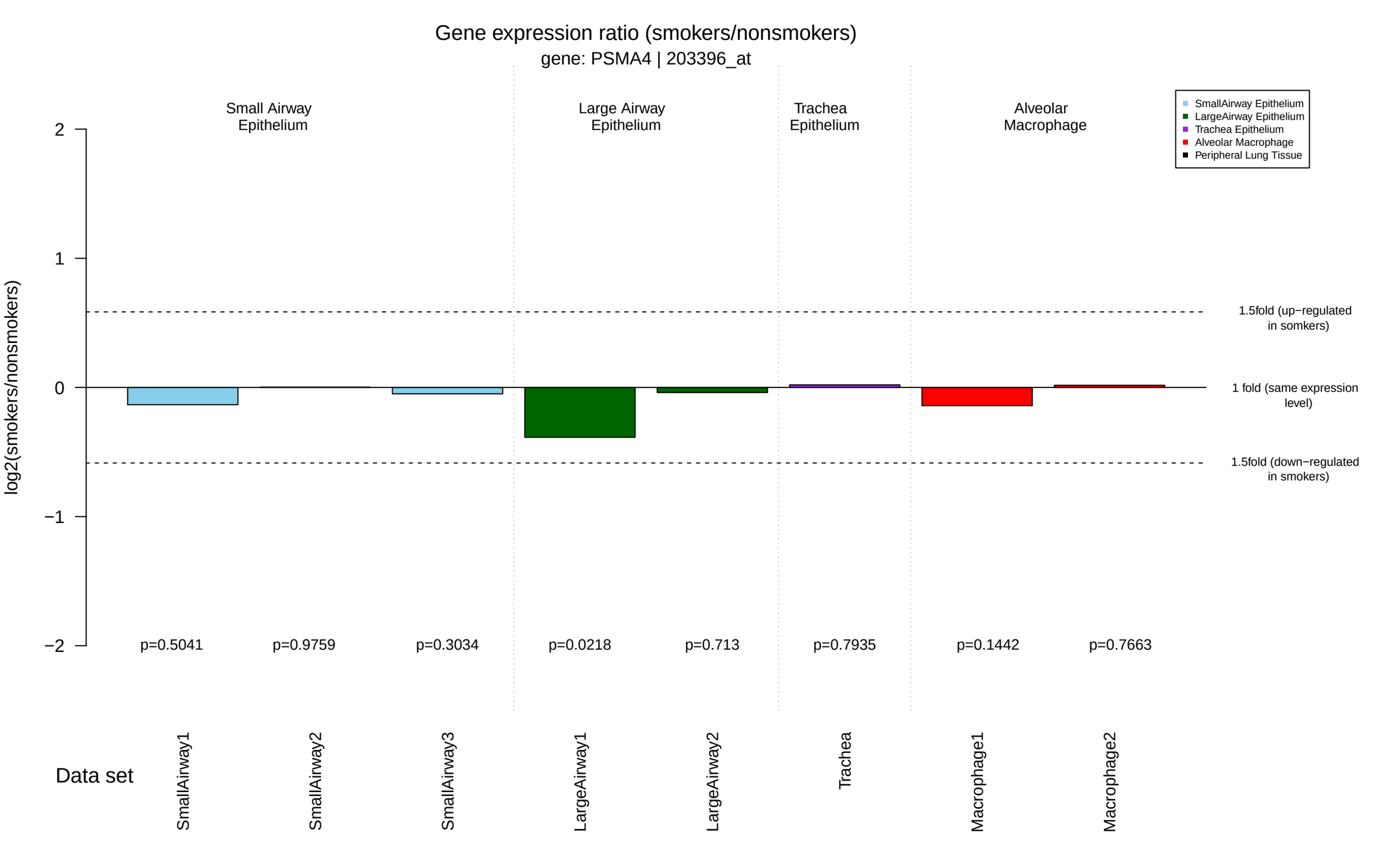
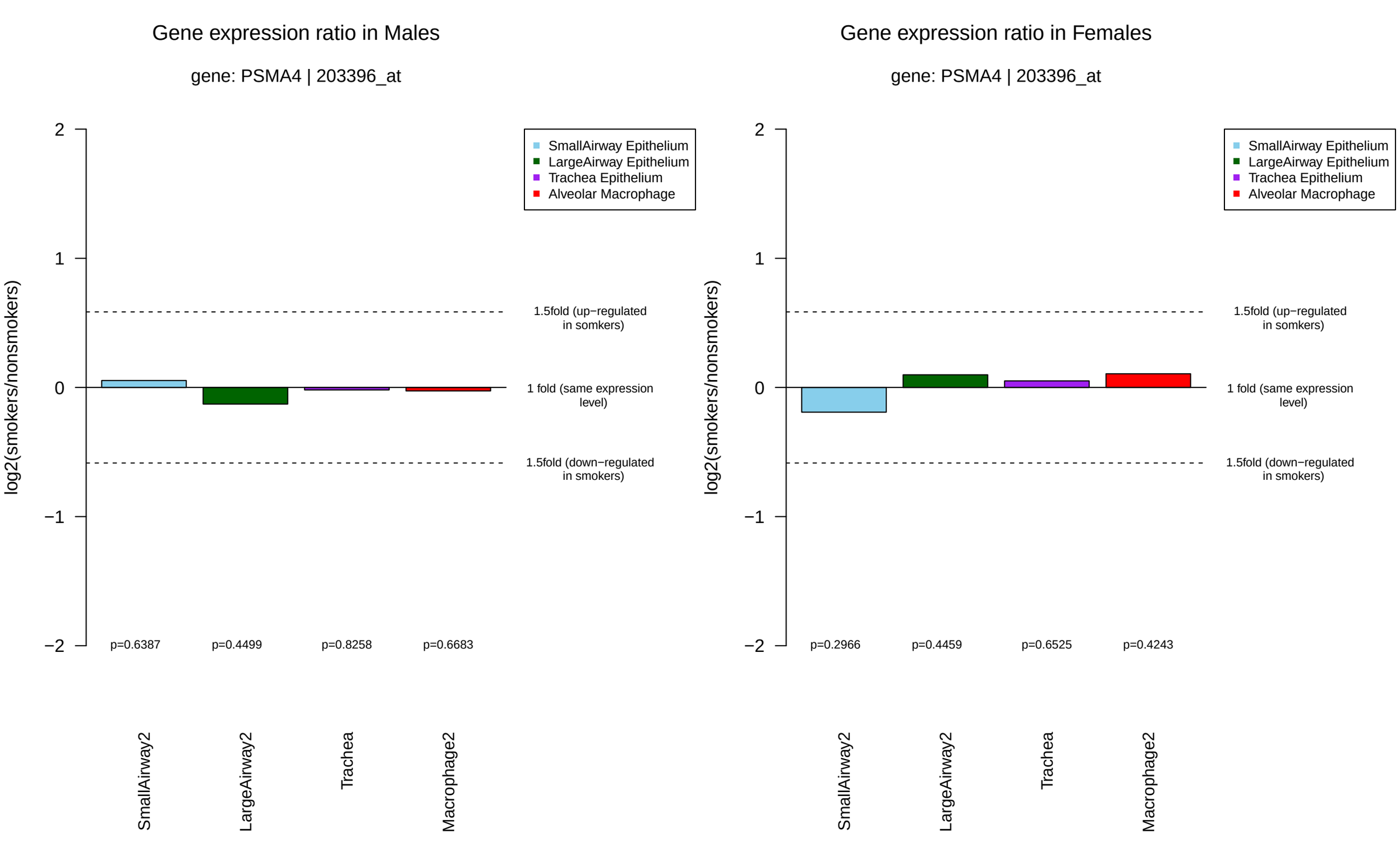
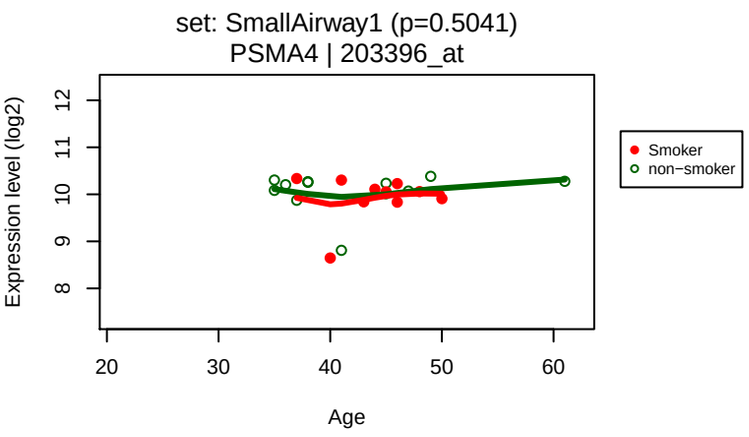
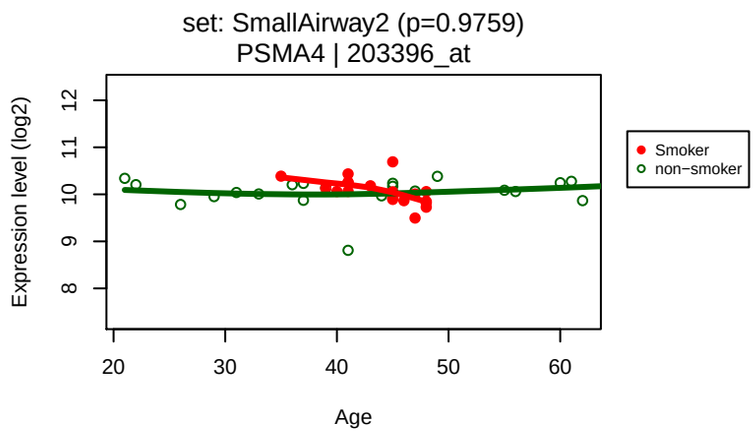
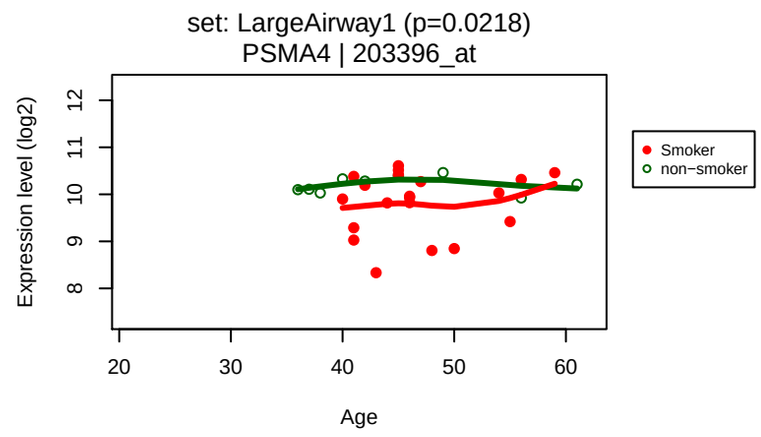

Gene expression of smokers and nonsmokers
gene: PSMA4 | 203396_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5041
p=0.9759
p=0.3034
p=0.0218
p=0.713
p=0.7935
p=0.1442
p=0.7663
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

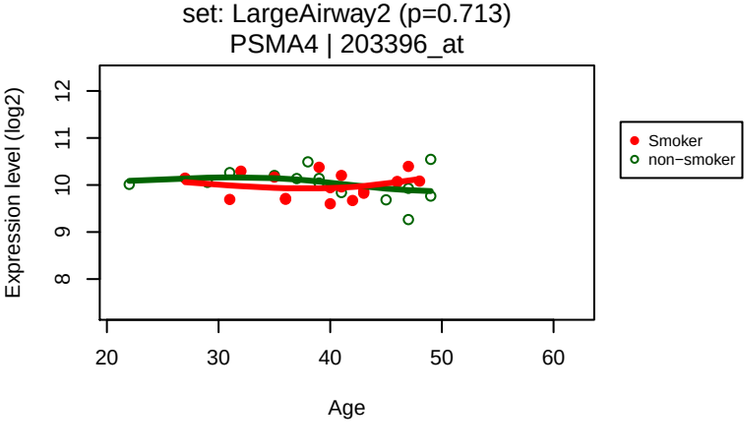

Gene expression ratio (smokers/nonsmokers)
gene: PSMA4 | 203396_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5041
p=0.9759
p=0.3034
p=0.0218
p=0.713
p=0.7935
p=0.1442
p=0.7663
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

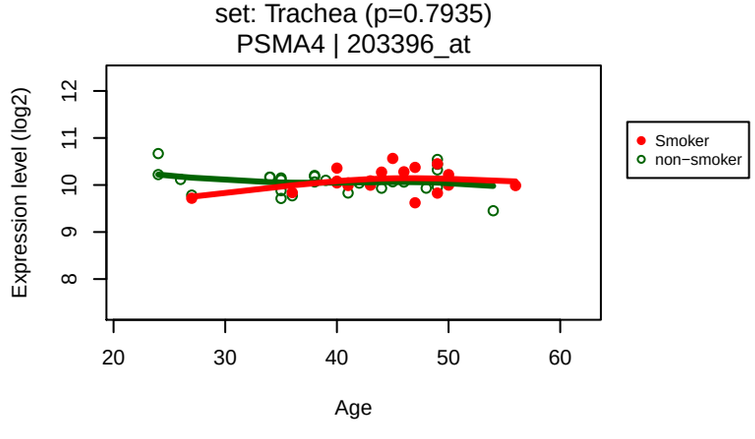

Gene expression ratio in Males
gene: PSMA4 | 203396_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6387
p=0.4499
p=0.8258
p=0.6683
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA4 | 203396_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2966
p=0.4459
p=0.6525
p=0.4243
SmallAirway2
LargeAirway2
Trachea
Macrophage2

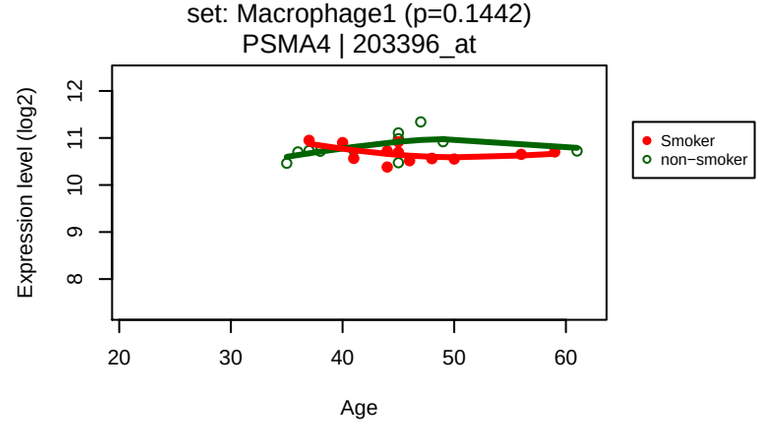

set: SmallAirway1 (p=0.5041)
PSMA4 | 203396_at
Expression level (log2)
Age
Smoker
non−smoker

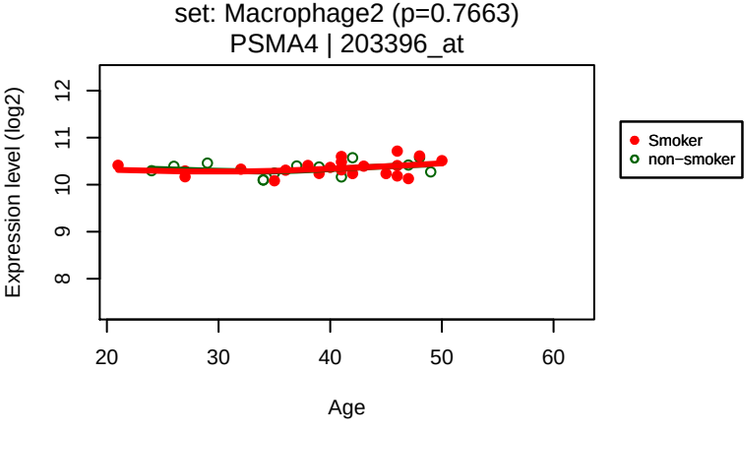

set: SmallAirway2 (p=0.9759)
PSMA4 | 203396_at
Expression level (log2)
Age
Smoker
non−smoker

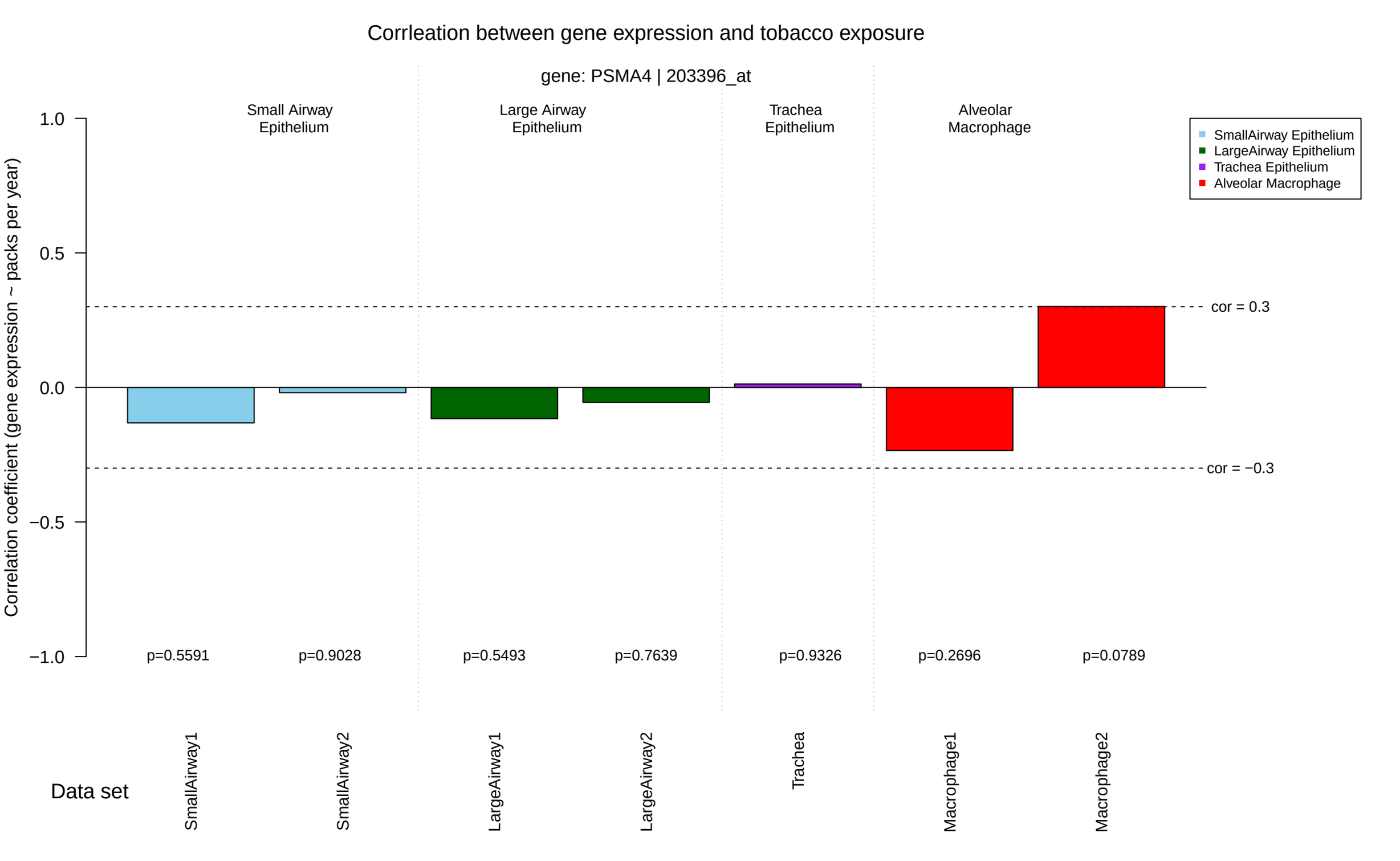

set: LargeAirway1 (p=0.0218)
PSMA4 | 203396_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.713)
PSMA4 | 203396_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7935)
PSMA4 | 203396_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1442)
PSMA4 | 203396_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7663)
PSMA4 | 203396_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA4 | 203396_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5591
p=0.9028
p=0.5493
p=0.7639
p=0.9326
p=0.2696
p=0.0789
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2